

Chapter 1

Necessary Fundamentals- Electromagnetic Fields and Waves

Reference Text Book: Field and Wave Electromagnetics by David K Cheng

Stated in a simple fashion, *electromagnetics* is the study of the effects of electric charges at rest and in motion. From elementary physics we know that there are two kinds of charges: positive and negative. Both positive and negative charges are sources of an electric field. Moving charges produce a current, which gives rise to a magnetic field. Here we tentatively speak of electric field and magnetic field in a general way; more definitive meanings will be attached to these terms later. A *field* is a spatial distribution of a quantity, which may or may not be a function of time. A time-varying electric field is accompanied by a magnetic field, and vice versa. In other words, time-varying electric and magnetic fields are coupled, resulting in an electromagnetic field. Under certain conditions, time-dependent electromagnetic fields produce waves that radiate from the source.

The concept of fields and waves is essential in the explanation of action at a distance. For instance, we learned from elementary mechanics that masses attract each other. This is why objects fall toward the earth's surface. But since there are no elastic strings connecting a free-falling object and the earth, how do we explain this phenomenon? We explain this action-at-a-distance phenomenon by postulating the existence of a gravitational field. The possibilities of satellite communication and of receiving signals from space probes millions of miles away can be explained only by postulating the existence of electric and magnetic fields and electromagnetic waves.

Electromagnetics is of fundamental importance to physicists and to electrical and computer engineers. Electromagnetic theory is indispensable in understanding the principle of atom smashers, cathode-ray oscilloscopes, radar, satellite communication, television reception, remote sensing, radio astronomy, microwave devices, optical fiber communication, transients in transmission lines, electromagnetic compatibility problems, instrument-landing systems, electromechanical energy conversion, and so on. Circuit concepts represent a restricted version, a special case, of electromagnetic concepts. As we shall see in the course, when the source frequency is very low so that the dimensions of a conducting network are much smaller than the wavelength, we have a quasi-static situation, which simplifies an electromagnetic problem to a circuit problem. However, we hasten to add that circuit theory is itself a highly developed, sophisticated discipline. It applies to a different class of electrical

engineering problems, and it is important in its own right.

Two situations illustrate the inadequacy of circuit-theory concepts and the need for electromagnetic-field concepts. Fig. 1.1 depicts a monopole antenna of the type we see on a walkie-talkie. On transmit, the source at the base feeds the antenna with a message-carrying current at an appropriate carrier frequency. From a circuit-theory point of view, the source feeds into an open circuit because the upper tip of the antenna is not connected to anything physically; hence no current would flow, and nothing would happen. This viewpoint, of course, cannot explain why communication can be established between walkie-talkies at a distance. Electromagnetic concepts must be used. We shall see in the course that when the length of the antenna is an appreciable part of the carrier wavelength, a nonuniform current will flow along the open-ended antenna. This current radiates a time-varying electromagnetic field in space, which propagates as an electromagnetic wave and induces currents in other antennas at a distance.



Figure 1.1: Monopole Antenna

Fig. 1.2 we show a situation in which an electromagnetic wave is incident from the left on a large conducting wall containing a small hole (aperture). Electromagnetic fields will exist on the right side of the wall at points, such as P in the figure, that are not necessarily directly behind the aperture. Circuit theory is obviously inadequate here for the determination (or even the explanation of the existence) of the field at P . The situation in Fig. 2, however, represents a problem of practical importance as its solution is relevant in evaluating the shielding effectiveness of the conducting wall.

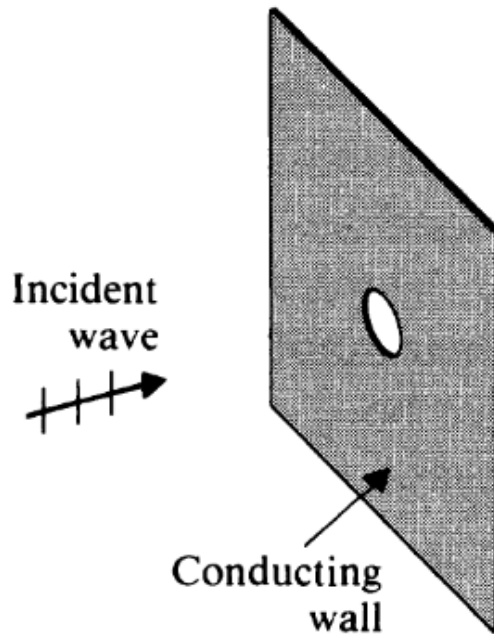


Figure 1.2: An Electromagnetic Problem

Generally speaking, circuit theory deals with lumped-parameter systems, circuits consisting of components characterized by lumped parameters such as resistances, inductances, and capacitances. Voltages and currents are the main system variables. For d-c circuits the system variables are constants, and the governing equations are algebraic equations. The system variables in a-c circuits are time-dependent; they are scalar quantities and are independent of space coordinates. The governing equations are ordinary differential equations. On the other hand, most electromagnetic variables are functions of time as well as of space coordinates. Many are vectors with both a magnitude and a direction, and their representation and manipulation require a knowledge of vector algebra and vector calculus. Even in static cases the governing equations are, in general, partial differential equations. It is essential that we be equipped to handle vector quantities and variables that are both time- and space-dependent. Techniques for solving partial differential equations are needed in dealing with certain types of electromagnetic problems. The importance of acquiring a facility in the use of these mathematical tools in the study of electromagnetics cannot be overemphasized.

Students who have mastered circuit theory may initially have the impression that electromagnetic theory is abstract. In fact, electromagnetic theory is no more abstract than circuit theory in the sense that the validity of both can be verified by experimentally measured results. In electromagnetics there is a need to define more quantities and to use more mathematical manipulations in order to develop a logical and complete theory that can explain a much wider variety of phenomena. The challenge of field and wave electromagnetics is not in the abstractness of the subject matter but rather in the process of mastering the electromagnetic model and the associated rules of operation. Dedication to acquiring this mastery will help us to meet the challenge and reap immeasurable satisfaction.

The electromagnetic theory can be built on a suitably chosen electromagnetic model. First step is defining the basic quantities of electromagnetics. The second step, the rules

of operation, encompasses vector algebra, vector calculus, and partial differential equations. The fundamentals of vector algebra and vector calculus. The third step, the fundamental postulates, will be presented in three substeps as we deal with static electric fields, steady magnetic fields, and electromagnetic fields, respectively.

The quantities in our electromagnetic model can be divided roughly into two categories: source quantities and field quantities. The source of an electromagnetic field is invariably electric charges at rest or in motion. However, an electromagnetic field may cause a redistribution of charges, which will, in turn, change the field; hence the separation between the cause and the effect is not always so distinct.

1.1 Vector Analysis

As we noted in earlier, some of the quantities in electromagnetics (such as charge, current, and energy) are scalars; and some others (such as electric and magnetic field intensities) are vectors. Both scalars and vectors can be functions of time and position. At a given time and position, a *scalar* is completely specified by its magnitude (positive or negative, together with its unit). Thus we can specify, for instance, a charge of $-1\ \mu\text{C}$ at a certain location at $t = 0$. The specification of a *vector* at a given location and time, on the other hand, requires both a magnitude and a direction. How do we specify the direction of a vector? In a three-dimensional space, three numbers are needed, and these numbers depend on the choice of a coordinate system. Conversion of a given vector from one coordinate system to another will change these numbers. However, physical laws and theorems relating various scalar and vector quantities certainly must hold irrespective of the coordinate system. The general expressions of the laws of electromagnetism, therefore, do not require the specification of a coordinate system. A particular coordinate system is chosen only when a problem of a given geometry is to be analyzed. For example, if we are to determine the magnetic field at the center of a current-carrying wire loop, it is more convenient to use rectangular coordinates if the loop is rectangular, whereas polar coordinates (two-dimensional) will be more appropriate if the loop is circular in shape. The basic electromagnetic relation governing the solution of such a problem is the same for both geometries.

Three main topics will be dealt with in this chapter on vector analysis:

1. Vector algebra—addition, subtraction, and multiplication of vectors.
2. Orthogonal coordinate systems—Cartesian, cylindrical, and spherical coordinates.
3. Vector calculus—differentiation and integration of vectors; line, surface, and volume integrals; “del” operator; gradient, divergence, and curl operations.

Vector Addition and Subtraction

We know that a vector has a magnitude and a direction. A vector $\mathbf{A}$ can be written as

$$\mathbf{A} = a_A \mathbf{A}, \quad (1.1)$$

where A is the magnitude (and has the unit and dimension) of $\mathbf{A}$,

$$A = |\mathbf{A}|, \quad (1.2)$$

and a_A is a dimensionless unit vector¹ with a unity magnitude having the direction of $\mathbf{A}$. Thus,

$$a_A = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\mathbf{A}}{A}. \quad (1.3)$$

The vector $\mathbf{A}$ can be represented graphically by a directed straight-line segment of length $|\mathbf{A}| = A$ with its arrowhead pointing in the direction of a_A , as shown in Fig. 1.3. Two vectors are equal if they have the same magnitude and the same direction, even though they may be displaced in space. Since it is difficult to write boldfaced letters by hand, it is a common practice to use an arrow or a bar over a letter ($\vec{A}$ or $\bar{A}$) or a wiggly line under a letter ($\tilde{A}$) to distinguish a vector from a scalar. This distinguishing mark, once chosen, *should never be omitted* whenever and wherever vectors are written.

Two vectors $\mathbf{A}$ and $\mathbf{B}$, which are not in the same direction nor in opposite directions, such as given in Fig. 1.4, determine a plane. Their sum is another vector $\mathbf{C}$ in the same plane. $\mathbf{C} = \mathbf{A} + \mathbf{B}$ can be obtained graphically in two ways.

1. By the parallelogram rule: The resultant $\mathbf{C}$ is the diagonal vector of the parallelogram formed by $\mathbf{A}$ and $\mathbf{B}$ drawn from the same point, as shown in Fig. 1.4.
2. By the head-to-tail rule: The head of $\mathbf{A}$ connects to the tail of $\mathbf{B}$. Their sum $\mathbf{C}$ is the vector drawn from the tail of $\mathbf{A}$ to the head of $\mathbf{B}$; and vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ form a triangle, as shown in Fig. 1.4.

It is obvious that vector addition obeys the commutative and associative laws.

$$\text{Commutative law: } \mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}, \quad (1.4)$$

$$\text{Associative law: } \mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}. \quad (1.5)$$

Vector subtraction can be defined in terms of vector addition in the following way:

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B}), \quad (1.6)$$

where $-\mathbf{B}$ is the negative of vector $\mathbf{B}$; that is, $-\mathbf{B}$ has the same magnitude as $\mathbf{B}$, but its direction is opposite to that of $\mathbf{B}$. Thus

$$-\mathbf{B} = (-a_B)\mathbf{B}. \quad (1.7)$$

The operation represented by Eq. (1-6) is illustrated in Fig. 1.5 illustrates vector subtraction..

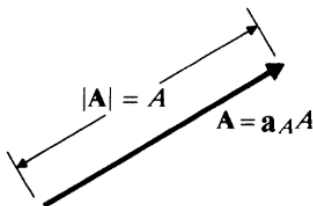


Figure 1.3: Graphical Representation of Vector A

¹A unit vector has a magnitude of unity.

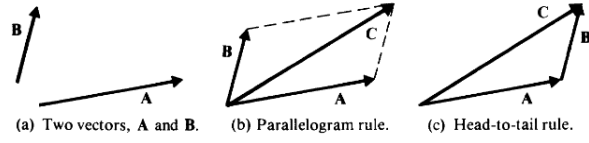


Figure 1.4: Vector Addition

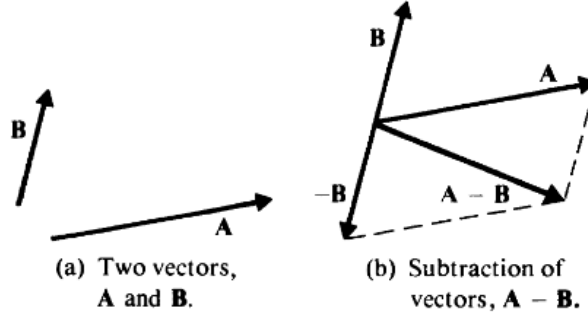


Figure 1.5: Vector Subtraction

Multiplication of a vector **A** by a positive scalar k changes the magnitude of **A** by k times without changing its direction (k can be either greater or less than 1).

$$k\mathbf{A} = a_A(kA). \quad (1.8)$$

It is not sufficient to say “the multiplication of one vector by another” or “the product of two vectors” because there are two distinct and very different types of products of two vectors. They are (1) scalar or dot products, and (2) vector or cross products. These will be defined in the following subsections.

Scalar or Dot Product

The scalar or dot product of two vectors **A** and **B**, denoted by $\mathbf{A} \cdot \mathbf{B}$, is a scalar, which equals the product of the magnitudes of **A** and **B** and the cosine of the angle between them. Thus,

$$\mathbf{A} \cdot \mathbf{B} \triangleq AB \cos \theta_{AB}. \quad (1.9)$$

In Eq. (1-9) the symbol $\triangleq$ signifies “equal by definition,” and θ_{AB} is the *smaller angle* between **A** and **B** and is less than π radians (180°), as indicated in Fig. 1.6. The dot product of two vectors (1) is less than or equal to the product of their magnitudes; (2) can be either a positive or a negative quantity, depending on whether the angle between them is smaller or larger than $\pi/2$ radians (90°); (3) is equal to the product

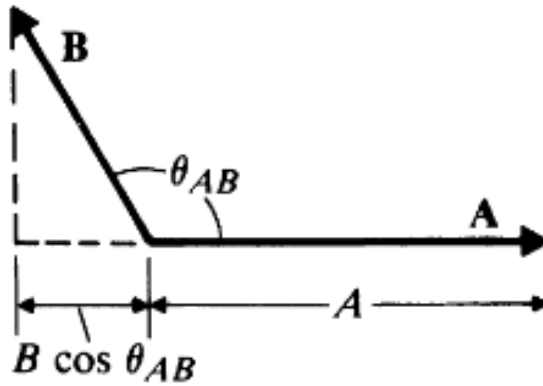


Figure 1.6: Illustrating the product of A and B

of the magnitude of one vector and the projection of the other vector upon the first one; and (4) is zero when the vectors are perpendicular to each other. It is evident that

$$\mathbf{A} \cdot \mathbf{A} = A^2 \quad (1.10)$$

or

$$A = \sqrt{\mathbf{A} \cdot \mathbf{A}}. \quad (1.11)$$

Equation (1-11) enables us to find the magnitude of a vector when the expression of the vector is given in any coordinate system.

The dot product is commutative and distributive.

$$\text{Commutative law: } \mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A} \quad (1.12)$$

$$\text{Distributive law: } \mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C} \quad (1.13)$$

The commutative law is obvious from the definition of the dot product in Eq. (1-9), and the proof of Eq. (1-13) is left as an exercise. The associative law does not apply to the dot product, since no more than two vectors can be so multiplied and an expression such as $\mathbf{A} \cdot \mathbf{B} \cdot \mathbf{C}$ is meaningless.

Example 2-1 Prove the law of cosines for a triangle.

Solution The law of cosines is a scalar relationship that expresses the length of a side of a triangle in terms of the lengths of the two other sides and the angle between them. Referring to Fig. 1.7, we find the law of cosines states that

$$C = \sqrt{A^2 + B^2 - 2AB \cos \alpha}$$

We prove this by considering the sides as vectors; that is,

$$\mathbf{C} = \mathbf{A} + \mathbf{B}$$

Taking the dot product of $\mathbf{C}$ with itself, we have, from Eqs. (1-10) and (1-13),

$$\begin{aligned} C^2 &= \mathbf{C} \cdot \mathbf{C} = (\mathbf{A} + \mathbf{B}) \cdot (\mathbf{A} + \mathbf{B}) \\ &= \mathbf{A} \cdot \mathbf{A} + \mathbf{B} \cdot \mathbf{B} + 2\mathbf{A} \cdot \mathbf{B} \\ &= A^2 + B^2 + 2AB \cos \theta_{AB} \end{aligned}$$

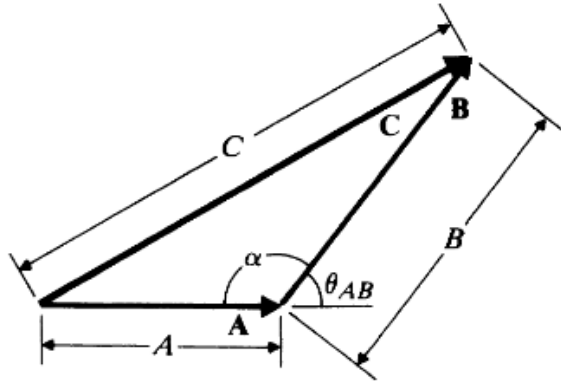


Figure 1.7: Example

Note that θ_{AB} is, by definition, the *smaller angle* between $\mathbf{A}$ and $\mathbf{B}$ and is equal to $(180^\circ - \alpha)$; hence

$$\cos \theta_{AB} = \cos(180^\circ - \alpha) = -\cos \alpha.$$

Therefore,

$$C^2 = A^2 + B^2 - 2AB \cos \alpha$$

and the law of cosines follows directly.

Vector or Cross Product

The vector or cross product of two vectors $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} \times \mathbf{B}$, is a vector perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$; its magnitude is $AB \sin \theta_{AB}$, where θ_{AB} is the *smaller angle* between $\mathbf{A}$ and $\mathbf{B}$, and its direction follows that of the thumb of the right hand when the fingers rotate from $\mathbf{A}$ to $\mathbf{B}$ through the angle θ_{AB} (the right-hand rule).

$$\mathbf{A} \times \mathbf{B} \triangleq a_n |AB \sin \theta_{AB}|. \quad (1.14)$$

This is illustrated in Fig. 1.8. Since $B \sin \theta_{AB}$ is the height of the parallelogram formed by the vectors $\mathbf{A}$ and $\mathbf{B}$, we recognize that the magnitude of $\mathbf{A} \times \mathbf{B}$, $|AB \sin \theta_{AB}|$, which is always positive, is numerically equal to the area of the parallelogram.

Using the definition in Eq. (1-14) and following the right-hand rule, we find that

$$\mathbf{B} \times \mathbf{A} = -\mathbf{A} \times \mathbf{B}. \quad (1.15)$$

Hence the cross product is *not* commutative. We can see that the cross product obeys the distributive law,

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}. \quad (1.16)$$

Can you show this in general without resolving the vectors into rectangular components?

The vector product is obviously *not* associative; that is,

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) \neq (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}. \quad (1.17)$$

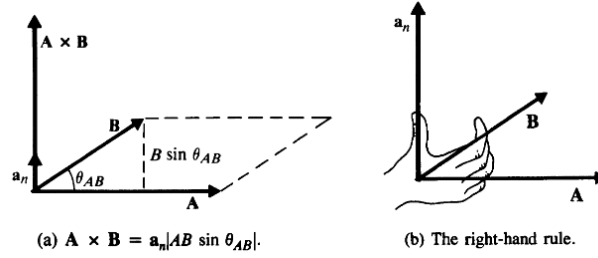


Figure 1.8: Cross Product of A and B

The vector representing the triple product on the left side of the expression above is perpendicular to $\mathbf{A}$ and lies in the plane formed by $\mathbf{B}$ and $\mathbf{C}$, whereas that on the right side is perpendicular to $\mathbf{C}$ and lies in the plane formed by $\mathbf{A}$ and $\mathbf{B}$. The order in which the two vector products are performed is therefore vital, and *in no case should the parentheses be omitted*.

Product of Three Vectors

There are two kinds of products of three vectors; namely, the *scalar triple product* and the *vector triple product*. The scalar triple product is much the simpler of the two and has the following property:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}). \quad (1.18)$$

Note the cyclic permutation of the order of the three vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$. Of course,

$$\begin{aligned} \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) &= -\mathbf{A} \cdot (\mathbf{C} \times \mathbf{B}) \\ &= -\mathbf{B} \cdot (\mathbf{A} \times \mathbf{C}) \\ &= -\mathbf{C} \cdot (\mathbf{B} \times \mathbf{A}) \end{aligned} \quad (1.19)$$

As can be seen from Fig. 1.9, each of the three expressions in Eq. (1-18) has a magnitude equal to the volume of the parallelepiped formed by the three vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$. The parallelepiped has a base with an area equal to $|\mathbf{B} \times \mathbf{C}| = |\mathbf{B} \mathbf{C} \sin \theta_1|$ and a height equal to $|\mathbf{A} \cos \theta_2|$; hence the volume is $|\mathbf{A} \mathbf{B} \mathbf{C} \sin \theta_1 \cos \theta_2|$.

The vector triple product $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ can be expanded as the difference of two simple vectors as follows:

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}). \quad (1.20)$$

Equation (1-20) is known as the “back-cab” rule and is a useful vector identity. (Note “BAC–CAB” on the right side of the equation!)

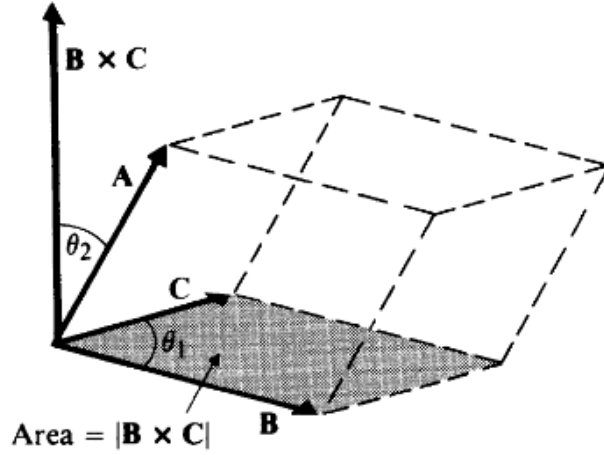


Figure 1.9: Illustrating scalar triple product

1.2 Orthogonal Coordinate Systems

We have indicated before that although the laws of electromagnetism are invariant with coordinate system, solution of practical problems requires that the relations derived from these laws be expressed in a coordinate system appropriate to the geometry of the given problems. For example, if we are to determine the electric field at a certain point in space, we at least need to describe the position of the source and the location of this point in a coordinate system. In a three-dimensional space a point can be located as the intersection of three surfaces. Assume that the three families of surfaces are described by $u_1 = \text{constant}$, $u_2 = \text{constant}$, and $u_3 = \text{constant}$, where the u 's need not all be lengths. (In the familiar Cartesian or rectangular coordinate system, u_1 , u_2 , and u_3 correspond to x , y , and z , respectively.) When these three surfaces are mutually perpendicular to one another, we have an *orthogonal coordinate system*. Nonorthogonal coordinate systems are not used because they complicate problems.

Some surfaces represented by $u_i = \text{constant}$ ($i = 1, 2, 3$) in a coordinate system may not be planes; they may be curved surfaces. Let $\mathbf{a}_{u_1}$, $\mathbf{a}_{u_2}$, and $\mathbf{a}_{u_3}$ be the unit vectors in the three coordinate directions. They are called the *base vectors*. In a general right-handed, orthogonal, curvilinear coordinate system the base vectors are arranged in such a way that the following relations are satisfied:

$$\begin{aligned}\mathbf{a}_{u_1} \times \mathbf{a}_{u_2} &= \mathbf{a}_{u_3}, \\ \mathbf{a}_{u_2} \times \mathbf{a}_{u_3} &= \mathbf{a}_{u_1}, \\ \mathbf{a}_{u_3} \times \mathbf{a}_{u_1} &= \mathbf{a}_{u_2}.\end{aligned}\tag{1.21}$$

These three equations are not all independent, as the specification of one automatically implies the other two. We have, of course,

$$\mathbf{a}_{u_1} \cdot \mathbf{a}_{u_2} = \mathbf{a}_{u_2} \cdot \mathbf{a}_{u_3} = \mathbf{a}_{u_3} \cdot \mathbf{a}_{u_1} = 0,\tag{1.22}$$

and

$$\mathbf{a}_{u_1} \cdot \mathbf{a}_{u_1} = \mathbf{a}_{u_2} \cdot \mathbf{a}_{u_2} = \mathbf{a}_{u_3} \cdot \mathbf{a}_{u_3} = 1.\tag{1.23}$$

Any vector $\mathbf{A}$ can be written as the sum of its components in the three orthogonal directions, as follows:

$$\mathbf{A} = \mathbf{a}_{u_1} A_{u_1} + \mathbf{a}_{u_2} A_{u_2} + \mathbf{a}_{u_3} A_{u_3}. \quad (1.24)$$

From Eq. (1-24) the magnitude of $\mathbf{A}$ is

$$A = |\mathbf{A}| = (A_{u_1}^2 + A_{u_2}^2 + A_{u_3}^2)^{1/2}. \quad (1.25)$$

In vector calculus (and in electromagnetics work) we are often required to perform line, surface, and volume integrals. In each case we need to express the differential length-change corresponding to a differential change in one of the coordinates. However, some of the coordinates, say u_i ($i = 1, 2, 3$), may not be a length; and a conversion factor is needed to convert a differential change du_i into a change in length $d\ell_i$:

$$d\ell_i = h_i du_i, \quad (1.26)$$

where h_i is called a *metric coefficient* and may itself be a function of u_1 , u_2 , and u_3 . For example, in the two-dimensional polar coordinates $(u_1, u_2) = (r, \phi)$, a differential change $d\phi (= du_2)$ in $\phi (= u_2)$ corresponds to a differential length-change $d\ell_2 = r d\phi$ ($h_2 = r = u_1$) in the $\mathbf{a}_\phi (= \mathbf{a}_{u_2})$ -direction. A directed differential length-change in an arbitrary direction can be written as the vector sum of the component length-changes:

$$d\boldsymbol{\ell} = \mathbf{a}_{u_1} d\ell_1 + \mathbf{a}_{u_2} d\ell_2 + \mathbf{a}_{u_3} d\ell_3 \quad (1.27)$$

or

$$d\boldsymbol{\ell} = \mathbf{a}_{u_1}(h_1 du_1) + \mathbf{a}_{u_2}(h_2 du_2) + \mathbf{a}_{u_3}(h_3 du_3). \quad (1.28)$$

In view of Eq. (1-25) the magnitude of $d\boldsymbol{\ell}$ is

$$d\ell = [(d\ell_1)^2 + (d\ell_2)^2 + (d\ell_3)^2]^{1/2} = [(h_1 du_1)^2 + (h_2 du_2)^2 + (h_3 du_3)^2]^{1/2}. \quad (1.29)$$

The differential volume dv formed by differential coordinate changes du_1 , du_2 , and du_3 in directions $\mathbf{a}_{u_1}$, $\mathbf{a}_{u_2}$, and $\mathbf{a}_{u_3}$, respectively, is $(d\ell_1 d\ell_2 d\ell_3)$, or

$$dv = h_1 h_2 h_3 du_1 du_2 du_3. \quad (1.30)$$

Later we will have occasion to express the current or flux flowing through a differential area. In such cases the cross-sectional area perpendicular to the current or flux flow must be used, and it is convenient to consider the differential area a vector with a direction normal to the surface; that is,

$$d\mathbf{s} = \mathbf{a}_n ds. \quad (1.31)$$

For instance, if current density $\mathbf{J}$ is not perpendicular to a differential area of magnitude ds , the current, dI , flowing through ds must be the component of $\mathbf{J}$ normal to the area multiplied by the area. Using the notation in Eq. (1-31), we can write simply

$$dI = \mathbf{J} \cdot d\mathbf{s} = \mathbf{J} \cdot \mathbf{a}_n ds. \quad (1.32)$$

In general orthogonal curvilinear coordinates the differential area ds_1 normal to the unit vector $\mathbf{a}_{u_1}$ is

$$ds_1 = d\ell_2 d\ell_3 \quad (1.33)$$

or

$$ds_1 = h_2 h_3 du_2 du_3. \quad (1.34)$$

Similarly, the differential areas normal to unit vectors $\mathbf{a}_{u_2}$ and $\mathbf{a}_{u_3}$ are, respectively,

$$ds_2 = h_1 h_3 du_1 du_3. \quad (1.35)$$

and

$$ds_3 = h_1 h_2 du_1 du_2. \quad (1.36)$$

Many orthogonal coordinate systems exist; but we shall be concerned only with the three that are most common and most useful:

1. Cartesian (or rectangular) coordinates.²
2. Cylindrical coordinates.
3. Spherical coordinates.

These will be discussed separately in the following subsections.

1.2.1 Cartesian Coordinates

$$(u_1, u_2, u_3) = (x, y, z)$$

A point $P(x_1, y_1, z_1)$ in Cartesian coordinates is the intersection of *three planes* specified by $x = x_1$, $y = y_1$, and $z = z_1$, as shown in Fig. 1–10. It is a right-handed system with base vectors $\mathbf{a}_x$, $\mathbf{a}_y$, and $\mathbf{a}_z$ satisfying the following relations:

$$\mathbf{a}_x \times \mathbf{a}_y = \mathbf{a}_z, \quad (1.37)$$

$$\mathbf{a}_y \times \mathbf{a}_z = \mathbf{a}_x, \quad (1.38)$$

$$\mathbf{a}_z \times \mathbf{a}_x = \mathbf{a}_y. \quad (1.39)$$

The position vector to the point $P(x_1, y_1, z_1)$ is

$$\mathbf{OP} = \mathbf{a}_x x_1 + \mathbf{a}_y y_1 + \mathbf{a}_z z_1. \quad (1.40)$$

A vector $\mathbf{A}$ in Cartesian coordinates can be written as

$$\mathbf{A} = \mathbf{a}_x A_x + \mathbf{a}_y A_y + \mathbf{a}_z A_z. \quad (1.41)$$

The dot product of two vectors $\mathbf{A}$ and $\mathbf{B}$ is, from Eq. (1–26),

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z. \quad (1.42)$$

and the cross product of $\mathbf{A}$ and $\mathbf{B}$ is, from Eq. (1–27),

$$\mathbf{A} \times \mathbf{B} = \mathbf{a}_x (A_y B_z - A_z B_y) + \mathbf{a}_y (A_z B_x - A_x B_z) + \mathbf{a}_z (A_x B_y - A_y B_x). \quad (1.43)$$

Since x , y , and z are lengths themselves, all three metric coefficients are unity; that is, $h_1 = h_2 = h_3 = 1$. The expressions for the differential length, differential area, and differential volume are—respectively,

²The term “Cartesian coordinates” is preferred because the term “rectangular coordinates” is customarily associated with two-dimensional geometry.

$$d\ell = \mathbf{a}_x dx + \mathbf{a}_y dy + \mathbf{a}_z dz. \quad (1.44)$$

$$ds_x = dy dz, \quad (1.45)$$

$$ds_y = dx dz, \quad (1.46)$$

$$ds_z = dx dy. \quad (1.47)$$

and

$$dv = dx dy dz. \quad (1.48)$$

A typical differential volume element at a point (x, y, z) resulting from differential changes dx , dy , and dz is shown in Fig. 1.10. The differential surface areas ds_x , ds_y , and ds_z normal to the directions $\mathbf{a}_x$, $\mathbf{a}_y$, and $\mathbf{a}_z$ are also indicated.

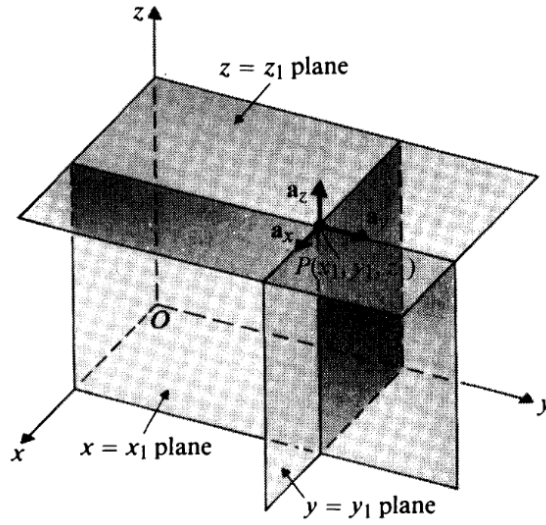


Figure 1.10: Cartesian Coordinates

1.2.2 Cylindrical Coordinates

$$(u_1, u_2, u_3) = (r, \phi, z)$$

In cylindrical coordinates a point $P(r_1, \phi_1, z_1)$ is the intersection of a circular cylindrical surface $r = r_1$, a half-plane containing the z -axis and making an angle $\phi = \phi_1$ with the xz -plane, and a plane parallel to the xy -plane at $z = z_1$. As indicated in Fig. 1.11, angle ϕ is measured from the positive x -axis, and the base vector $\mathbf{a}_\phi$ is

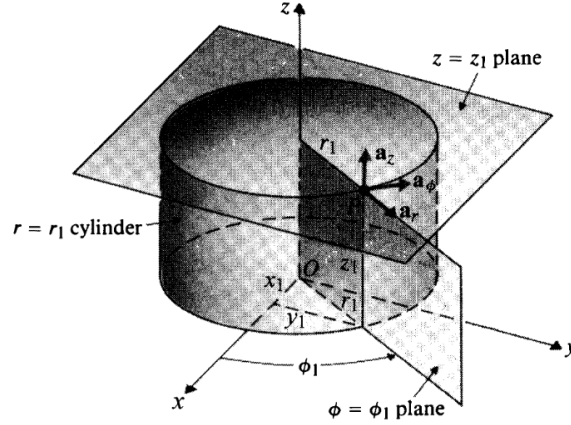


Figure 1.11: Cylindrical Coordinates

tangential to the cylindrical surface. The following right-hand relations apply:

$$\begin{aligned}\mathbf{a}_r \times \mathbf{a}_\phi &= \mathbf{a}_z, \\ \mathbf{a}_\phi \times \mathbf{a}_z &= \mathbf{a}_r\end{aligned}\tag{1.49}$$

$$\mathbf{a}_z \times \mathbf{a}_r = \mathbf{a}_\phi.\tag{1.50}$$

Cylindrical coordinates are important for problems with long line charges or currents, and in places where cylindrical or circular boundaries exist. The two-dimensional polar coordinates are a special case at $z = 0$.

A vector in cylindrical coordinates is written as

$$\mathbf{A} = \mathbf{a}_r A_r + \mathbf{a}_\phi A_\phi + \mathbf{a}_z A_z.\tag{1.51}$$

The expressions for the dot and cross products of two vectors in cylindrical coordinates follow from Eqs. (1–26) and (2–27) directly.

Two of the three coordinates, r and z (u_1, u_3), are themselves lengths; hence $h_1 = h_3 = 1$. However, ϕ is an angle requiring a metric coefficient $h_2 = r$ to convert $d\phi$ to $d\ell_2$. The general expression for a differential length in cylindrical coordinates is then, from Eq. (1–31),

$$d\ell = \mathbf{a}_r dr + \mathbf{a}_\phi r d\phi + \mathbf{a}_z dz.\tag{1.52}$$

The expressions for differential areas and differential volume are

$$\begin{aligned}ds_r &= r d\phi dz, \\ ds_\phi &= dr dz, \\ ds_z &= r dr d\phi,\end{aligned}\tag{1.53}$$

and

$$dv = r dr d\phi dz.\tag{1.54}$$

A typical differential volume element at a point (r, ϕ, z) resulting from differential changes dr , $d\phi$, and dz in the three orthogonal coordinate directions is shown in Fig. 1.12.

A vector given in cylindrical coordinates can be transformed into one in Cartesian coordinates, and vice versa. Suppose we want to express $\mathbf{A} = \mathbf{a}_r A_r + \mathbf{a}_\phi A_\phi + \mathbf{a}_z A_z$ in Cartesian

coordinates; that is, we want to write $\mathbf{A}$ as $A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$ and determine A_x , A_y , and A_z . First of all, we note that A_z , the z -component of $\mathbf{A}$, is not changed by the transformation from cylindrical to Cartesian coordinates. To find A_x , we equate the dot products of both expressions of $\mathbf{A}$ with $\mathbf{a}_x$. Thus $A_x = \mathbf{A} \cdot \mathbf{a}_x$
 $= A_r \mathbf{a}_r \cdot \mathbf{a}_x + A_\phi \mathbf{a}_\phi \cdot \mathbf{a}_x$

The term containing A_z disappears here because $\mathbf{a}_z \cdot \mathbf{a}_x = 0$. Referring to Fig. 1.13, which shows the relative positions of the base vectors $\mathbf{a}_x$, $\mathbf{a}_y$, $\mathbf{a}_r$, and $\mathbf{a}_\phi$, we see that

$$\mathbf{a}_r \cdot \mathbf{a}_x = \cos \phi, \quad (1.55)$$

and

$$\mathbf{a}_\phi \cdot \mathbf{a}_x = \cos\left(\frac{\pi}{2} + \phi\right) = -\sin \phi. \quad (1.56)$$

Hence,

$$A_x = A_r \cos \phi - A_\phi \sin \phi. \quad (1.57)$$

Similarly, to find A_y , we take the dot products of both expressions of $\mathbf{A}$ with $\mathbf{a}_y$: $A_y = \mathbf{A} \cdot \mathbf{a}_y$
 $= A_r \mathbf{a}_r \cdot \mathbf{a}_y + A_\phi \mathbf{a}_\phi \cdot \mathbf{a}_y$

From Fig. 1.13 we find that

$$\mathbf{a}_r \cdot \mathbf{a}_y = \cos\left(\frac{\pi}{2} - \phi\right) = \sin \phi, \quad (1.58)$$

and

$$\mathbf{a}_\phi \cdot \mathbf{a}_y = \cos \phi. \quad (1.59)$$

It follows that

$$A_y = A_r \sin \phi + A_\phi \cos \phi. \quad (1.60)$$

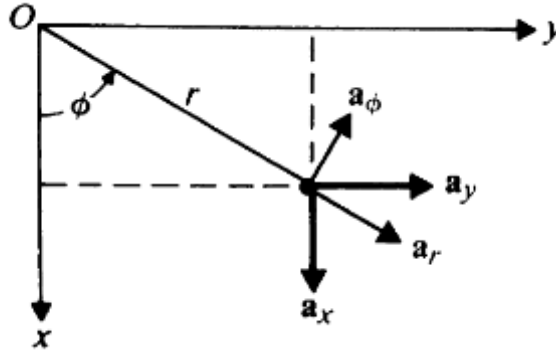


Figure 1.13: Relations between $\mathbf{a}_x$, $\mathbf{a}_y$, $\mathbf{a}_r$, and $\mathbf{a}_\phi$.

It is convenient to write the relations between the components of a vector in Cartesian and cylindrical coordinates in a matrix form:

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix}. \quad (1.61)$$

Our problem is now solved except that the $\cos \phi$ and $\sin \phi$ in Eq. (1-61) should be converted into Cartesian coordinates. Moreover, A_r , A_ϕ , and A_z may themselves be functions of r , ϕ , and z . In that case, they too should be converted into functions of x , y , and z in the final answer. The following conversion formulas are obvious from Fig. 1.13. From cylindrical to Cartesian coordinates:

$$\begin{aligned}x &= r \cos \phi, \\y &= r \sin \phi, \\z &= z.\end{aligned}\tag{1.62}$$

The inverse relations (from Cartesian to cylindrical coordinates) are

$$\begin{aligned}r &= \sqrt{x^2 + y^2}, \\ \phi &= \tan^{-1}\left(\frac{y}{x}\right), \\ z &= z.\end{aligned}\tag{1.63}$$

1.2.3 Spherical Coordinates

$$(u_1, u_2, u_3) = (R, \theta, \phi)$$

A point $P(R_1, \theta_1, \phi_1)$ in spherical coordinates is specified as the intersection of the following three surfaces: a spherical surface centered at the origin with a radius $R = R_1$; a right circular cone with its apex at the origin, its axis coinciding with the z -axis and having a half-angle $\theta = \theta_1$; and a half-plane containing the z -axis and making an angle $\phi = \phi_1$ with the xz -plane. The base vector $\mathbf{a}_R$ at P is radial from the origin and is quite different from $\mathbf{a}_r$ in cylindrical coordinates, the latter being perpendicular to the z -axis. The base vector $\mathbf{a}_\theta$ lies in the $\phi = \phi_1$ plane and is tangential to the

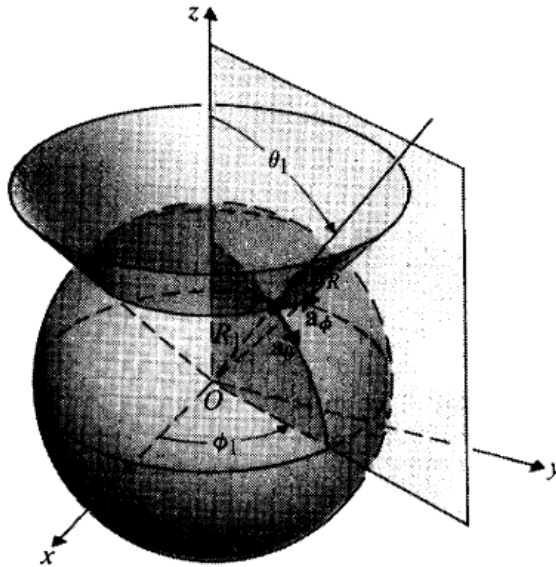


Figure 1.14: Spherical coordinates

spherical surface, whereas the base vector $\mathbf{a}_\phi$ is the same as that in cylindrical coordinates. These are illustrated in Fig. 1.14. For a right-handed system we have

$$\begin{aligned}\mathbf{a}_R \times \mathbf{a}_\theta &= \mathbf{a}_\phi, \\ \mathbf{a}_\theta \times \mathbf{a}_\phi &= \mathbf{a}_R, \\ \mathbf{a}_\phi \times \mathbf{a}_R &= \mathbf{a}_\theta.\end{aligned}\tag{1.64}$$

Spherical coordinates are important for problems involving point sources and regions with spherical boundaries. When an observer is very far from the source region of a finite extent, the latter could be considered as the origin of a spherical coordinate system; and, as a result, suitable simplifying approximations could be made. This is the reason that spherical coordinates are used in solving antenna problems in the far field.

A vector in spherical coordinates is written as

$$\mathbf{A} = \mathbf{a}_R A_R + \mathbf{a}_\theta A_\theta + \mathbf{a}_\phi A_\phi.\tag{1.65}$$

The expressions for the dot and cross products of two vectors in spherical coordinates can be obtained from Eqs. (1-26) and (2-27).

In spherical coordinates, only R (u_1) is a length. The other two coordinates, θ and ϕ (u_2 and u_3), are angles. Referring to Fig. 1.15, in which a typical differential volume element is shown, we see that metric coefficients $h_2 = R$ and $h_3 = R \sin \theta$ are required to convert $d\theta$ and $d\phi$ into $d\ell_2$ and $d\ell_3$, respectively. The general expression for a differential length is, from Eq. (1-31),

$$d\ell = \mathbf{a}_R dR + \mathbf{a}_\theta R d\theta + \mathbf{a}_\phi R \sin \theta d\phi.\tag{1.66}$$

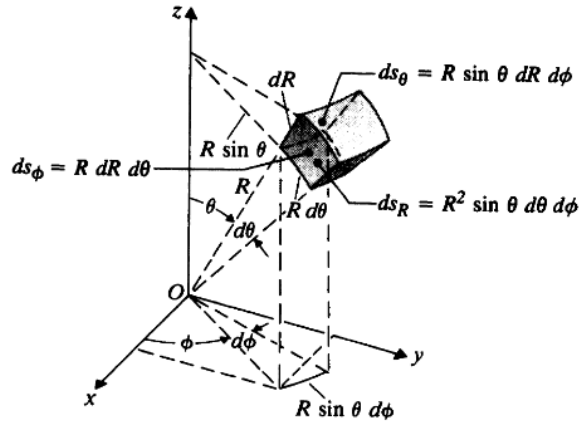


Figure 1.15: A differential volume element in spherical coordinates

The expressions for differential areas and differential volume resulting from differential changes dR , $d\theta$, and $d\phi$ in the three coordinate directions are

$$\begin{aligned}ds_R &= R^2 \sin \theta d\theta d\phi, \\ ds_\theta &= R \sin \theta dR d\phi, \\ ds_\phi &= R dR d\theta,\end{aligned}\tag{1.67}$$

and

$$dv = R^2 \sin \theta dR d\theta d\phi.\tag{1.68}$$

For convenience the base vectors, metric coefficients, and expressions for the differential volume are tabulated in Fig 1.16.

Three Basic Orthogonal Coordinate Systems

Coordinate System Relations		Cartesian Coordinates (x, y, z)	Cylindrical Coordinates (r, ϕ, z)	Spherical Coordinates (R, θ, ϕ)
Base vectors	$\mathbf{a}_{u_1}$	$\mathbf{a}_x$	$\mathbf{a}_r$	$\mathbf{a}_R$
	$\mathbf{a}_{u_2}$	$\mathbf{a}_y$	$\mathbf{a}_\phi$	$\mathbf{a}_\theta$
	$\mathbf{a}_{u_3}$	$\mathbf{a}_z$	$\mathbf{a}_z$	$\mathbf{a}_\phi$
Metric coefficients	h_1	1	1	1
	h_2	1	r	R
	h_3	1	1	$R \sin \theta$
Differential volume	dv	$dx dy dz$	$r dr d\phi dz$	$R^2 \sin \theta dR d\theta d\phi$

Figure 1.16: Comparison Table

A vector given in spherical coordinates can be transformed into one in Cartesian or cylindrical coordinates, and vice versa. From Fig. 1.16 it is easily seen that

$$\begin{aligned}
 x &= R \sin \theta \cos \phi, \\
 y &= R \sin \theta \sin \phi, \\
 z &= R \cos \theta.
 \end{aligned}
 \tag{1.69}$$

Conversely, measurements in Cartesian coordinates can be transformed into those in spherical coordinates:

$$\begin{aligned}
 R &= \sqrt{x^2 + y^2 + z^2}, \\
 \theta &= \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right), \\
 \phi &= \tan^{-1} \left(\frac{y}{x} \right).
 \end{aligned}
 \tag{1.70}$$

1.3 Integrals Containing Vector Functions

In electromagnetics work we have occasion to encounter integrals that contain vector functions such as

$$\int_V \mathbf{F} dv, \quad (1.71)$$

$$\int_C V d\ell, \quad (1.72)$$

$$\int_C \mathbf{F} \cdot d\ell, \quad (1.73)$$

$$\int_S \mathbf{A} \cdot d\mathbf{s}. \quad (1.74)$$

The volume integral in (1.71) can be evaluated as the sum of three scalar integrals by first resolving the vector $\mathbf{F}$ into its three components in the appropriate coordinate system. If dv denotes a differential volume, then (1.71) is actually a shorthand way of representing a triple integral over three dimensions.

In the second integral, in (1.72), V is a scalar function of space, $d\ell$ represents a differential increment of length, and C is the path of integration. If the integration is to be carried out from a point P_1 to another point P_2 , we write $\int_{P_1}^{P_2} V d\ell$. If the integration is to be evaluated around a closed path C , we denote it by $\oint_C V d\ell$. In Cartesian coordinates, (1.72) can be written as

$$\int_C V d\ell = \int_C V(x, y, z) [\mathbf{a}_x dx + \mathbf{a}_y dy + \mathbf{a}_z dz], \quad (1.75)$$

in view of Eq. (1-44). Since the Cartesian unit vectors are constant in both magnitude and direction, they can be taken out of the integral sign, and Eq. (1-82) becomes

$$\int_C V d\ell = \mathbf{a}_x \int_C V(x, y, z) dx + \mathbf{a}_y \int_C V(x, y, z) dy + \mathbf{a}_z \int_C V(x, y, z) dz. \quad (1.76)$$

The three integrals on the right-hand side of Eq. (1.76) are ordinary scalar integrals; they can be evaluated for a given $V(x, y, z)$ around a path C .

The expression in (1.74), $\int_S \mathbf{A} \cdot d\mathbf{s}$, is a surface integral. It is actually a double integral over two dimensions; but it is written with a single integral sign for simplicity. The integral measures the flux of the vector field $\mathbf{A}$ flowing through the area S . In the integral the vector differential surface element $d\mathbf{s} = \mathbf{a}_n ds$ has a magnitude ds and a direction denoted by the unit vector $\mathbf{a}_n$. The conventions for the positive direction of ds or $\mathbf{a}_n$ are as follows:

1. If the surface of integration, S , is a *closed surface* enclosing a volume, then the positive direction for $\mathbf{a}_n$ is always in the outward direction from the volume. This is illustrated in Fig. 1.17(a). We see that the positive direction of $\mathbf{a}_n$ depends on the location of ds . A small circle is added over the integral sign if the integration is to be performed over an enclosed surface:

$$\oint_S \mathbf{A} \cdot d\mathbf{s} = \oint_S \mathbf{A} \cdot \mathbf{a}_n ds. \quad (1.77)$$

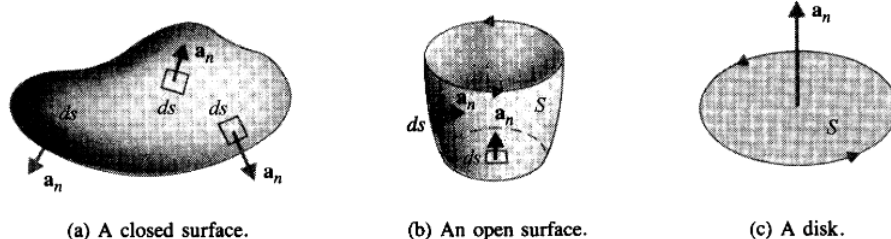


Figure 1.17: Illustrating the positive direction of $\mathbf{a}_n$ in scalar surface integral

If S is an open surface, the positive direction for $\mathbf{a}_n$ depends on the direction in which the perimeter of the open surface is traversed. This is illustrated in Fig. 1.17(b), in which a cup-shaped surface (with no lid) is shown. We apply the right-hand rule: If the fingers of the right hand follow the direction of travel around the perimeter, then the thumb points in the direction of positive $\mathbf{a}_n$. Here again, the positive direction of $\mathbf{a}_n$ depends on the location of ds . A plane, such as the disk in Fig. 1.17(c), is a special case of an open surface where $\mathbf{a}_n$ is a constant.

1.4 Gradient of a Scalar Field

In electromagnetics we have to deal with quantities that depend on both time and position. Since three coordinate variables are involved in a three-dimensional space, we expect to encounter scalar and vector fields that are functions of four variables: (t, u_1, u_2, u_3) . In general, the fields may change as any one of the four variables changes. We now address the method for describing the space rate of change of a scalar field at a given time. Partial derivatives with respect to the three space-coordinate variables are involved, and, inasmuch as the rate of change may be different in different directions, a vector is needed to define the space rate of change of a scalar field at a given point and at a given time.

Let us consider a scalar function of space coordinates $V(u_1, u_2, u_3)$, which may represent, say, the temperature distribution in a building, the altitude of a mountainous terrain, or the electric potential in a region. The magnitude of V , in general, depends on the position of the point in space, but it may be constant along certain lines or surfaces. Figure 1.18 shows two surfaces on which the magnitude of V is constant and has the values V_1 and $V_1 + dV$, respectively, where dV indicates a small change in V . We should note that constant- V surfaces need not coincide with any of the surfaces that define a particular coordinate system. Point P_1 is on surface V_1 ; P_2 is the corresponding point on surface $V_1 + dV$ along the normal vector $d\mathbf{n}$; and P_3 is a point close to P_2 along another vector $d\boldsymbol{\ell} \neq d\mathbf{n}$. For the same change dV in V , the space rate of change, $dV/d\ell$, is obviously greatest along $d\mathbf{n}$ because $d\mathbf{n}$ is the

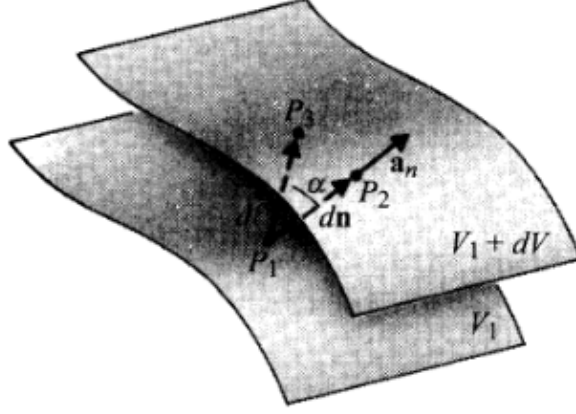


Figure 1.18: Gradient of a scalar

shortest distance between the two surfaces. Since the magnitude of $dV/d\ell$ depends on the direction of $d\ell$, $dV/d\ell$ is a directional derivative. We define the vector that represents both the magnitude and the direction of the maximum space rate of increase of a scalar as the gradient of that scalar. We write

$$\text{grad } V \triangleq \mathbf{a}_n \frac{dV}{dn}. \quad (1.78)$$

For brevity it is customary to employ the operator *del*, represented by the symbol ∇ , and write ∇V in place of $\text{grad } V$. Thus,

$$\nabla V \triangleq \mathbf{a}_n \frac{dV}{dn}. \quad (1.79)$$

We have assumed that dV is positive (an increase in V); if dV is negative (a decrease in V from P_1 to P_2), ∇V will be negative in the $\mathbf{a}_n$ direction.

The directional derivative along $d\ell$ is

$$\begin{aligned} \frac{dV}{d\ell} &= \frac{dV}{dn} \frac{dn}{d\ell} = \frac{dV}{dn} \cos \alpha \\ &= \frac{dV}{dn} \mathbf{a}_n \cdot \mathbf{a}_\ell = (\nabla V) \cdot \mathbf{a}_\ell \end{aligned} \quad (1.80)$$

Equation (1-80) states that the space rate of increase of V in the $\mathbf{a}_\ell$ direction is equal to the projection (the component) of the gradient of V in that direction. We can also write Eq. (1-80) as

$$dV = (\nabla V) \cdot d\ell, \quad (1.81)$$

where $d\ell = \mathbf{a}_\ell d\ell$.

Now, dV in Eq. (1-81) is the total differential of V as a result of a change in position (from P_1 to P_3 in Fig. 1.18); it can be expressed in terms of the differential changes in u_1 , u_2 , and u_3 :

$$dV = \frac{\partial V}{\partial \ell_1} d\ell_1 + \frac{\partial V}{\partial \ell_2} d\ell_2 + \frac{\partial V}{\partial \ell_3} d\ell_3, \quad (1.82)$$

where $d\ell_1$, $d\ell_2$, and $d\ell_3$ are the components of the vector differential displacement $d\ell$ in a chosen coordinate system. In terms of general orthogonal curvilinear coordinates (u_1, u_2, u_3) , $d\ell$ is (from Eq. 1-31)

$$d\ell = \mathbf{a}_{u_1}d\ell_1 + \mathbf{a}_{u_2}d\ell_2 + \mathbf{a}_{u_3}d\ell_3 = \mathbf{a}_{u_1}(h_1du_1) + \mathbf{a}_{u_2}(h_2du_2) + \mathbf{a}_{u_3}(h_3du_3). \quad (1.83)$$

We can write dV in Eq. (1-82) as the dot product of two vectors, as follows:

$$\begin{aligned} dV &= \left(\mathbf{a}_{u_1} \frac{\partial V}{\partial \ell_1} + \mathbf{a}_{u_2} \frac{\partial V}{\partial \ell_2} + \mathbf{a}_{u_3} \frac{\partial V}{\partial \ell_3} \right) \cdot (\mathbf{a}_{u_1}d\ell_1 + \mathbf{a}_{u_2}d\ell_2 + \mathbf{a}_{u_3}d\ell_3) \\ &= \left(\mathbf{a}_{u_1} \frac{\partial V}{\partial \ell_1} + \mathbf{a}_{u_2} \frac{\partial V}{\partial \ell_2} + \mathbf{a}_{u_3} \frac{\partial V}{\partial \ell_3} \right) \cdot d\ell \end{aligned}$$

Comparing Eq. (1-81) with above equation, we obtain

$$\nabla V = \mathbf{a}_{u_1} \frac{\partial V}{\partial \ell_1} + \mathbf{a}_{u_2} \frac{\partial V}{\partial \ell_2} + \mathbf{a}_{u_3} \frac{\partial V}{\partial \ell_3}. \quad (1.84)$$

or

$$\nabla V = \mathbf{a}_{u_1} \frac{1}{h_1} \frac{\partial V}{\partial u_1} + \mathbf{a}_{u_2} \frac{1}{h_2} \frac{\partial V}{\partial u_2} + \mathbf{a}_{u_3} \frac{1}{h_3} \frac{\partial V}{\partial u_3}. \quad (1.85)$$

Equation (1-85) is a useful formula for computing the gradient of a scalar when the scalar is given as a function of space coordinates.

In Cartesian coordinates, $(u_1, u_2, u_3) = (x, y, z)$, and $h_1 = h_2 = h_3 = 1$, we have

$$\nabla V = \mathbf{a}_x \frac{\partial V}{\partial x} + \mathbf{a}_y \frac{\partial V}{\partial y} + \mathbf{a}_z \frac{\partial V}{\partial z}. \quad (1.86)$$

or

$$\nabla V = \left(\mathbf{a}_x \frac{\partial}{\partial x} + \mathbf{a}_y \frac{\partial}{\partial y} + \mathbf{a}_z \frac{\partial}{\partial z} \right) V. \quad (1.87)$$

In view of Eq. (1-87), it is convenient to consider ∇ in Cartesian coordinates as a vector differential operator:

$$\nabla = \mathbf{a}_x \frac{\partial}{\partial x} + \mathbf{a}_y \frac{\partial}{\partial y} + \mathbf{a}_z \frac{\partial}{\partial z}. \quad (1.88)$$

From Eq. (1-85), we see that we can define ∇ as

$$\nabla = \mathbf{a}_{u_1} \frac{1}{h_1} \frac{\partial}{\partial u_1} + \mathbf{a}_{u_2} \frac{1}{h_2} \frac{\partial}{\partial u_2} + \mathbf{a}_{u_3} \frac{1}{h_3} \frac{\partial}{\partial u_3}. \quad (1.89)$$

in general orthogonal curvilinear coordinates. As we shall see later in this chapter, the same vector differential operator is also used to signify divergence ($\nabla \cdot$) and curl ($\nabla \times$) operations on a vector. In these cases it is important to remember that the differentiation of a base vector in a curvilinear coordinate system may lead to a new vector in a different direction. Proper care must be exercised when ∇ is used to operate on vectors in curvilinear coordinate systems.

1.5 Divergence of a Vector Field

In the preceding section we considered the spatial derivatives of a scalar field, which led to the definition of the gradient. We now turn our attention to the spatial derivatives of a vector field. This will lead to the definitions of the divergence and the curl of a vector. Both are very important in the study of electromagnetism.

In the study of vector fields it is convenient to represent field variations graphically by directed field lines, which are called *flux lines* or *streamlines*. They are directed lines or curves that indicate at each point the direction of the vector field, as illustrated in Fig. 1.19. The magnitude of the field at a point is depicted either by the density or by the length of the directed lines in the vicinity of the point. Figure 1.19(a) shows that the field in region *A* is stronger than that in region *B* because there is a higher density of equally long directed lines in region *A*. In Fig. 1.19(b), the decreasing arrow lengths away from the point *q* indicate a radial field that is strongest in the region closest to *q*. Figure 1.19(c) depicts a uniform field.

The vector field strength in Fig. 1.19(a) is measured by the number of flux lines passing through a unit surface normal to the vector. The flux of a vector field is analogous to the flow of an incompressible fluid such as water. For a volume with an enclosed surface there will be an excess outward or inward flow through the surface only when the volume contains a source or a sink, respectively; that is, a net positive divergence indicates the presence of a source of fluid inside the volume, and a net negative divergence indicates the presence of a sink. The net outward flow of the fluid per unit volume is therefore a measure of the strength of the enclosed source. In the uniform field shown in Fig. 1.19(c) there is an equal amount of inward and outward flux going through any closed volume containing no sources or sinks, resulting in a zero divergence.

We define the divergence of a vector field $\mathbf{A}$ at a point, abbreviated $\text{div } \mathbf{A}$, as the net outward flux of $\mathbf{A}$ per unit volume as the volume about the point tends to zero:

$$\text{div } \mathbf{A} \triangleq \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{s}}{\Delta v}. \quad (1.90)$$

The numerator in Eq. (1-90), representing the net outward flux, is an integral over the entire surface *S* that bounds the volume. Equation (1.90) is the general definition of $\text{div } \mathbf{A}$, which is a scalar quantity whose magnitude may vary from point to point as $\mathbf{A}$ itself varies. This definition holds for any coordinate system; the expression for $\text{div } \mathbf{A}$, like that for $\mathbf{A}$, will, of course, depend on the choice of the coordinate system.

At the beginning of this section we intimated that the divergence of a vector is a type of spatial derivative. The reader might perhaps wonder about the presence of an integral in the expression given by Eq. (1-90); but a two-dimensional surface integral divided by a three-dimensional volume will lead to spatial derivatives as the volume approaches zero. We shall now derive the expression for $\text{div } \mathbf{A}$ in Cartesian coordinates.

Consider a differential volume of sides Δx , Δy , and Δz centered about a point $P(x_0, y_0, z_0)$ in the field of a vector $\mathbf{A}$, as shown in Fig. 1.20. In Cartesian coordinates,

$$\mathbf{A} = a_x \mathbf{A}_x + a_y \mathbf{A}_y + a_z \mathbf{A}_z.$$

We wish to find $\text{div } \mathbf{A}$ at the point (x_0, y_0, z_0) . Since the differential volume has six faces, the surface integral in the numerator of Eq. (1-90) can be decomposed into six parts:

$$\oint_S \mathbf{A} \cdot d\mathbf{s} = \left[\int_{\text{front face}} + \int_{\text{back face}} + \int_{\text{right face}} + \int_{\text{left face}} + \int_{\text{top face}} + \int_{\text{bottom face}} \right] \mathbf{A} \cdot d\mathbf{s}. \quad (1.91)$$

On the front face,

$$\begin{aligned} \int_{\text{front face}} \mathbf{A} \cdot d\mathbf{s} &= A_{\text{front face}} \cdot \Delta s_{\text{front face}} = A_{\text{front face}} \cdot a_x(\Delta y \Delta z) \\ &= A_x \left(x_0 + \frac{\Delta x}{2}, y_0, z_0 \right) \Delta y \Delta z. \end{aligned} \quad (1.92)$$

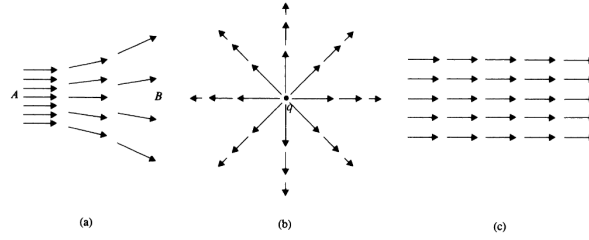


Figure 1.19: Flux lines of vector fields

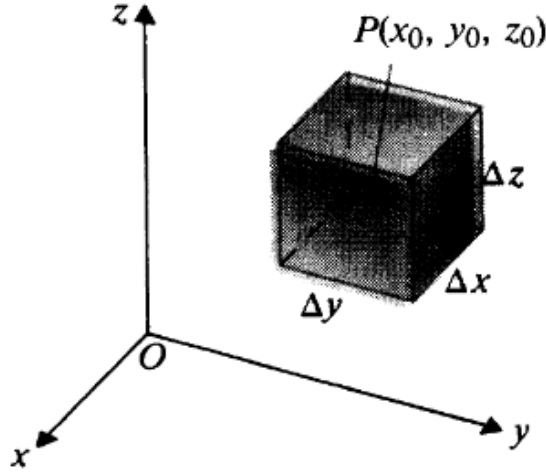


Figure 1.20: A differential volume in cartesian coordinates

The quantity $A_x(x_0 + \frac{\Delta x}{2}, y_0, z_0)$ can be expanded as a Taylor series about its value at (x_0, y_0, z_0) as follows:

$$A_x \left(x_0 + \frac{\Delta x}{2}, y_0, z_0 \right) = A_x(x_0, y_0, z_0) + \frac{\Delta x}{2} \frac{\partial A_x}{\partial x} \bigg|_{(x_0, y_0, z_0)} + \text{higher-order terms}, \quad (1.93)$$

where the higher-order terms (H.O.T.) contain the factors $(\Delta x)^2$, $(\Delta x)^3$, etc.

Similarly, on the back face,

$$\int_{\text{back face}} \mathbf{A} \cdot d\mathbf{s} = A_{\text{back face}} \cdot \Delta s_{\text{back face}} = A_{\text{back face}} \cdot (-a_x \Delta y \Delta z) \quad (1.94)$$

$$= -A_x \left(x_0 - \frac{\Delta x}{2}, y_0, z_0 \right) \Delta y \Delta z.$$

The Taylor-series expansion of $A_x(x_0 - \frac{\Delta x}{2}, y_0, z_0)$ is

$$A_x \left(x_0 - \frac{\Delta x}{2}, y_0, z_0 \right) = A_x(x_0, y_0, z_0) - \frac{\Delta x}{2} \frac{\partial A_x}{\partial x} \Big|_{(x_0, y_0, z_0)} + \text{H.O.T.} \quad (1.95)$$

Substituting Eqs. (1.93) in Eq. (1.92) and Eq. (1.95) in Eq. (1.94) and adding the contributions, we have

$$\left[\int_{\text{front face}} + \int_{\text{back face}} \right] \mathbf{A} \cdot d\mathbf{s} = \left(\frac{\partial A_x}{\partial x} + \text{H.O.T.} \right)_{(x_0, y_0, z_0)} \Delta x \Delta y \Delta z. \quad (1.96)$$

Here a Δx has been factored out from the H.O.T. in Eqs. (1.93) and (1.94), but all terms of the H.O.T. in Eq. (1.96) still contain powers of Δx .

Following the same procedure for the right and left faces, where the coordinate changes are $+\Delta y/2$ and $-\Delta y/2$, respectively, and $\Delta s = \Delta x \Delta y$, we find

$$\left[\int_{\text{right face}} + \int_{\text{left face}} \right] \mathbf{A} \cdot d\mathbf{s} = \left(\frac{\partial A_y}{\partial y} + \text{H.O.T.} \right)_{(x_0, y_0, z_0)} \Delta x \Delta y \Delta z. \quad (1.97)$$

Here the higher-order terms contain the factors Δy , $(\Delta y)^2$, etc. For the top and bottom faces we have

$$\left[\int_{\text{top face}} + \int_{\text{bottom face}} \right] \mathbf{A} \cdot d\mathbf{s} = \left(\frac{\partial A_z}{\partial z} + \text{H.O.T.} \right)_{(x_0, y_0, z_0)} \Delta x \Delta y \Delta z, \quad (1.98)$$

where the higher-order terms contain the factors Δz , $(\Delta z)^2$, etc.

Now the results from Eqs. (1.96), (1.97), and (1.98) are combined in Eq. (1.91) to obtain

$$\oint_S \mathbf{A} \cdot d\mathbf{s} = \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right)_{(x_0, y_0, z_0)} \Delta x \Delta y \Delta z + \text{higher-order terms in } \Delta x, \Delta y, \Delta z. \quad (1.99)$$

Since $\Delta v = \Delta x \Delta y \Delta z$, substitution of Eq. (1.99) in Eq. (1-90) yields the expression of $\text{div } \mathbf{A}$ in Cartesian coordinates:

$$\text{div } \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}. \quad (1.100)$$

The higher-order terms vanish as the differential volume $\Delta x \Delta y \Delta z$ approaches zero. The value of $\text{div } \mathbf{A}$, in general, depends on the position of the point at which it is evaluated. We have dropped the notation (x_0, y_0, z_0) in Eq. (1-108) because it applies to any point at which $\mathbf{A}$ and its partial derivatives are defined.

With the vector differential operator del , ∇ , defined in Eq. (1-88), we can write Eq. (1-100) alternatively as $\nabla \cdot \mathbf{A}$; that is,

$$\nabla \cdot \mathbf{A} = \text{div } \mathbf{A}. \quad (1.101)$$

In general orthogonal curvilinear coordinates (u_1, u_2, u_3) , Eq. (1-90) will lead to

$$\nabla \cdot \mathbf{A} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (h_2 h_3 A_1) + \frac{\partial}{\partial u_2} (h_1 h_3 A_2) + \frac{\partial}{\partial u_3} (h_1 h_2 A_3) \right]. \quad (1.102)$$

1.6 Divergence Theorem

In the preceding section we defined the divergence of a vector field as the net outward flux per unit volume. We may expect intuitively that the *volume integral of the divergence of a vector field equals the total outward flux of the vector through the surface that bounds the volume*; that is,

$$\int_V \nabla \cdot \mathbf{A} \, dv = \oint_S \mathbf{A} \cdot d\mathbf{s} \quad (1.103)$$

This identity, which will be proved in the following paragraph, is called the *divergence theorem*. It applies to any volume V that is bounded by surface S . The direction of $d\mathbf{s}$ is always that of the outward normal, perpendicular to the surface and directed away from the volume.

For a very small differential volume element Δv_j bounded by a surface s_j , the definition of $\nabla \cdot \mathbf{A}$ in Eq. (1.90) gives directly

$$(\nabla \cdot \mathbf{A})_j \Delta v_j = \oint_{s_j} \mathbf{A} \cdot d\mathbf{s} \quad (1.104)$$

In case of an arbitrary volume V , we can subdivide it into many, say N , small differential volumes, of which Δv_j is typical. This is depicted in Fig. 1.21. Let us now combine the contributions of all these differential volumes to both sides of Eq. (1.104). We have

$$\lim_{\Delta v_j \rightarrow 0} \left[\sum_{j=1}^N (\nabla \cdot \mathbf{A})_j \Delta v_j \right] = \lim_{\Delta v_j \rightarrow 0} \left[\sum_{j=1}^N \oint_{s_j} \mathbf{A} \cdot d\mathbf{s} \right] \quad (1.105)$$

The left side of Eq. (1.105) is, by definition, the volume integral of $\nabla \cdot \mathbf{A}$:

$$\lim_{\Delta v_j \rightarrow 0} \sum_{j=1}^N (\nabla \cdot \mathbf{A})_j \Delta v_j = \int_V (\nabla \cdot \mathbf{A}) \, dv \quad (1.106)$$

The surface integrals on the right side of Eq. (1.105) are summed over all the faces of all the differential volume elements. The contributions from the internal surfaces of adjacent elements will, however, cancel each other, because at a common internal surface the outward normals of the adjacent elements point in opposite directions. Hence the net contribution of the right side of Eq. (1.105) is due only to that of the external surface S bounding the volume V ; that is,

$$\lim_{\Delta v_j \rightarrow 0} \left[\sum_{j=1}^N \oint_{s_j} \mathbf{A} \cdot d\mathbf{s} \right] = \oint_S \mathbf{A} \cdot d\mathbf{s} \quad (1.107)$$

The substitution of Eqs. (1.106) and (1.107) in Eq. (1.105) yields the divergence theorem in Eq. (1.103).

The validity of the limiting processes leading to the proof of the divergence theorem requires that the vector field $\mathbf{A}$, as well as its first derivatives, exist and be continuous both in V and on S . The divergence theorem is an important identity in vector analysis. It converts a volume integral of the divergence of a vector to a closed surface integral of the

vector, and vice versa. We use it frequently in establishing other theorems and relations in electromagnetics. We emphasize that, although a single integral sign is used on both sides of Eq. (1.103) for simplicity, the volume and surface integrals represent triple and double integrations, respectively.

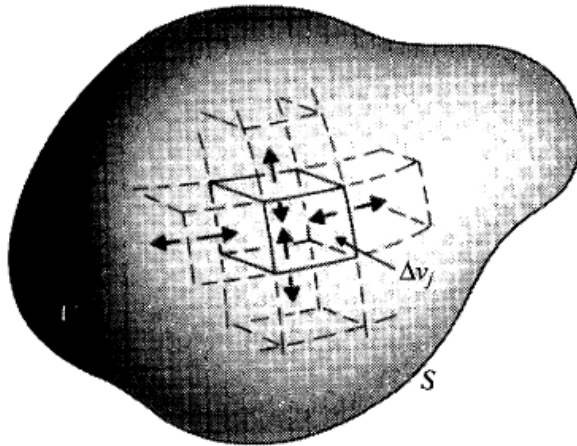


Figure 1.21: Subdivide volume for proof of Divergence theorem

1.7 Curl of a Vector Field

In the last section, we stated that a net outward flux of a vector $\mathbf{A}$ through a surface bounding a volume indicates the presence of a source. This source may be called a flow source, and $\nabla \cdot \mathbf{A}$ is a measure of the strength of the flow source. There is another kind of source, called a vortex source, which causes a circulation of a vector field around it. The net circulation (or simply circulation) of a vector field around a closed path is defined as the scalar line integral of the vector over the path. We have

$$\text{Circulation of } \mathbf{A} \text{ around contour } C = \oint_C \mathbf{A} \cdot d\boldsymbol{\ell} \quad (1.108)$$

Equation (1.108) is a mathematical definition. The physical meaning of circulation depends on what kind of field the vector $\mathbf{A}$ represents. If $\mathbf{A}$ is a force acting on an object, its circulation will be the work done by the force in moving the object once around the contour; if $\mathbf{A}$ represents an electric field intensity, then the circulation will be an electromotive force around the closed path. The familiar phenomenon of water whirling down a sink drain is an example of a vortex sink causing a circulation of fluid velocity. A circulation of $\mathbf{A}$ may exist even when $\nabla \cdot \mathbf{A} = 0$ (when there is no flow source).

Since circulation as defined in Eq. (1.108) is a line integral of a dot product, its value obviously depends on the orientation of the contour C relative to the vector $\mathbf{A}$. In order to define a point function, which is a measure of the strength of a vortex source, we must make C very small and orient it in such a way that the circulation is a maximum. We define

$$\text{curl } \mathbf{A} \equiv \nabla \times \mathbf{A} = \lim_{\Delta s \rightarrow 0} \frac{1}{\Delta s} \left[\mathbf{a}_n \oint_C \mathbf{A} \cdot d\boldsymbol{\ell} \right]_{\max} \quad (1.109)$$

In words, Eq. (1.109) states that the curl of a vector field $\mathbf{A}$, denoted by $\text{curl } \mathbf{A}$ or $\nabla \times \mathbf{A}$, is a vector whose magnitude is the maximum net circulation of $\mathbf{A}$ per unit area as the area tends to zero and whose direction is the normal direction of the area when the area is oriented to make the net circulation maximum. Because the normal to an area can point in two opposite directions, we adhere to the right-hand rule.

The component of $\nabla \times \mathbf{A}$ in any direction $\mathbf{a}_u$ is

$$(\nabla \times \mathbf{A})_u = \mathbf{a}_u \cdot (\nabla \times \mathbf{A}) = \lim_{\Delta s_u \rightarrow 0} \frac{1}{\Delta s_u} \left(\oint_{C_u} \mathbf{A} \cdot d\boldsymbol{\ell} \right) \quad (1.110)$$

We now use Eq. (1.110) to find the three components of $\nabla \times \mathbf{A}$ in Cartesian coordinates. Refer to Fig. 1.22. We have $\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$.

After evaluating the contributions of the contour integrals and taking the limit, the x -component of the curl is obtained as

$$(\nabla \times \mathbf{A})_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \quad (1.111)$$

Similarly, the full expression for the curl in Cartesian coordinates is

$$\nabla \times \mathbf{A} = \mathbf{a}_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \mathbf{a}_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \mathbf{a}_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \quad (1.112)$$

This can also be written in determinant form:

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \quad (1.113)$$

In general orthogonal curvilinear coordinates (u_1, u_2, u_3) , the curl is given by

$$\nabla \times \mathbf{A} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \mathbf{a}_1 & h_2 \mathbf{a}_2 & h_3 \mathbf{a}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 A_1 & h_2 A_2 & h_3 A_3 \end{vmatrix} \quad (1.114)$$

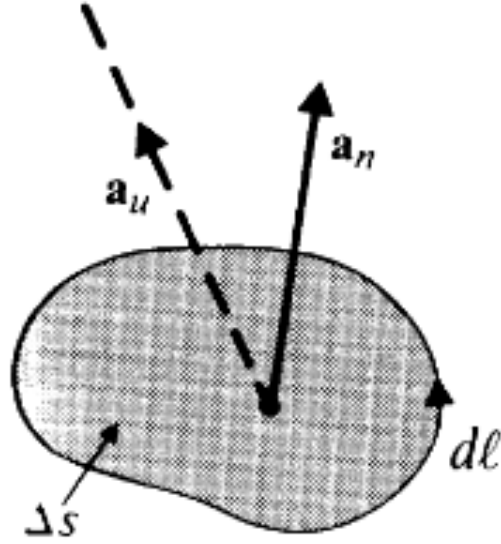


Figure 1.22: Relation between $\mathbf{a}_n$ and $d\ell$ in defining curl

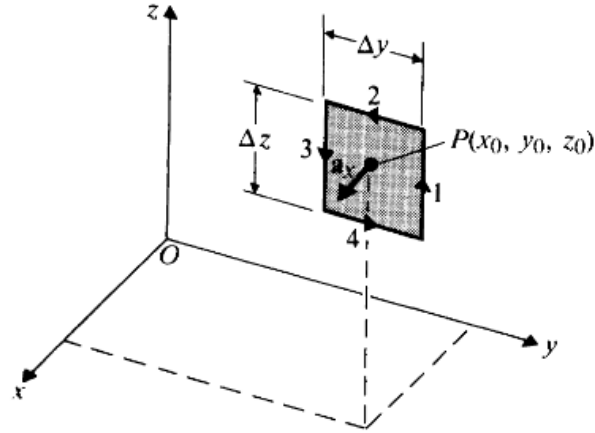


Figure 1.23: Determining $(\nabla \times \mathbf{A})_x$

1.8 Stokes's Theorem

For a very small differential area Δs_j bounded by a contour C_j , the definition of $\nabla \times \mathbf{A}$ in Eq. (1.109) leads to

$$(\nabla \times \mathbf{A})_j \cdot (\Delta s_j) = \oint_{C_j} \mathbf{A} \cdot d\ell \quad (1.115)$$

In obtaining Eq. (1.115), we have taken the dot product of both sides of Eq. (1.109) with $\mathbf{a}_n \Delta s_j$ or Δs_j . For an arbitrary surface S , we can subdivide it into many, say N , small differential areas. Figure 1.23 shows such a scheme with Δs_j as a typical differential element.

The left side of Eq. (1.115) is the flux of the vector $\nabla \times \mathbf{A}$ through the area Δs_j . Adding the contributions of all the differential areas to the flux, we have

$$\lim_{\Delta s_j \rightarrow 0} \sum_{j=1}^N (\nabla \times \mathbf{A})_j \cdot (\Delta s_j) = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} \quad (1.116)$$

Now we sum up the line integrals around the contours of all the differential elements represented by the right side of Eq. (1.115). Since the common part of the contours of two adjacent elements is traversed in opposite directions by two contours, the net contribution of all the common parts in the interior to the total line integral is zero, and only the contribution from the external contour C bounding the entire area S remains after the summation:

$$\lim_{\Delta s_j \rightarrow 0} \sum_{j=1}^N \left(\oint_{C_j} \mathbf{A} \cdot d\boldsymbol{\ell} \right) = \oint_C \mathbf{A} \cdot d\boldsymbol{\ell} \quad (1.117)$$

Combining Eqs. (1.116) and (1.117), we obtain Stokes's theorem:

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\boldsymbol{\ell} \quad (1.118)$$

which states that the surface integral of the curl of a vector field over an open surface is equal to the closed line integral of the vector along the contour bounding the surface.

As with the divergence theorem, the validity of the limiting processes leading to Stokes's theorem requires that the vector field $\mathbf{A}$, as well as its first derivatives, exist and be continuous both on S and along C . Stokes's theorem converts a surface integral of the curl of a vector to a line integral of the vector, and vice versa.

If the surface integral of $\nabla \times \mathbf{A}$ is carried over a closed surface, there will be no surface-bounding external contour, and Eq. (1.118) tells us that

$$\oint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = 0 \quad (1.119)$$

for any closed surface S . The geometry in Fig. 1-24 is chosen deliberately to emphasize the fact that a nontrivial application of Stokes's theorem always implies an open surface with a rim. The simplest open surface would be a two-dimensional plane or disk with its circumference as the contour. We remind ourselves here that the directions of $d\boldsymbol{\ell}$ and $d\mathbf{s}$ follow the right-hand rule.

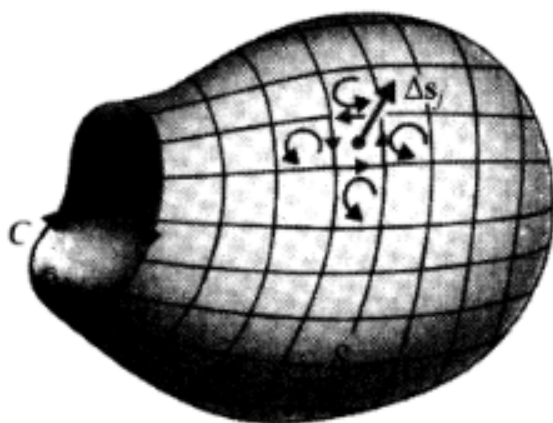


Figure 1.24: Subdivided area for proof of stokes theorem